MAT 5210 Homework 3 — Solutions

Problem 1

Let $\Phi_m(x) = \prod_{\zeta \text{ primitive } m\text{-th root of } 1}(x-\zeta) \in \mathbb{C}[x]$.

(a) The primitive m-th roots of unity for small m are: $m=1$: 1; $m=2$: -1; $m=3$: ω, ω^2 where $\omega = e^{2\pi i/3}$; $m=4$: $\pm i$. Hence

$$\Phi_1 = x-1, \qquad \Phi_2 = x+1,$$
$$\Phi_3 = (x-\omega)(x-\omega^2) = x^2 - (\omega+\omega^2)x + \omega^3 = x^2+x+1, \quad \Phi_4 = (x-i)(x+i) = x^2+1.$$

(b) The polynomial x^m-1 has m distinct roots in $\mathbb{C}$ (its derivative mx^{m-1} has only the root 0), namely all ζ with $\zeta^m = 1$, so $x^m - 1 = \prod_{\zeta^m=1}(x-\zeta)$. Each such ζ has a multiplicative order d, and $d \mid m$; then ζ is a primitive d-th root of unity. Conversely, for each $d \mid m$ every primitive d-th root of unity satisfies $\zeta^m = 1$. An element has only one order, so the roots of x^m-1 are partitioned according to $d \mid m$ into the sets of primitive d-th roots, and

$$x^m - 1 = \prod_{d \mid m} \prod_{\zeta \text{ primitive } d\text{-th}} (x-\zeta) = \prod_{d \mid m} \Phi_d(x).$$

(c) We show by strong induction on m that $\Phi_m \in \mathbb{Z}[x]$ (and it is monic by definition). For $m=1$, $\Phi_1 = x-1 \in \mathbb{Z}[x]$. Let $m > 1$ and assume $\Phi_d \in \mathbb{Z}[x]$ for all $d < m$. Put $g = \prod_{d \mid m,\, d<m} \Phi_d \in \mathbb{Z}[x]$, a monic polynomial. By (b), $x^m - 1 = \Phi_m \cdot g$ in $\mathbb{C}[x]$. Since g is monic, division with remainder in $\mathbb{Z}[x]$ gives $x^m - 1 = qg + r$ with $q, r \in \mathbb{Z}[x]$ and $\deg r < \deg g$. By uniqueness of division with remainder in $\mathbb{C}[x]$, $q = \Phi_m$ and $r = 0$. Hence $\Phi_m \in \mathbb{Z}[x]$. (In particular $\Phi_m \in \mathbb{Q}[x]$.)

(d) Let p be prime. The divisors of p are $1, p$, so by (b) $\Phi_1\Phi_p = x^p - 1$, and

$$\Phi_p = \frac{x^p-1}{x-1} = 1 + x + \cdots + x^{p-1}.$$

For $n \geq 1$ the divisors of p^n are $1, p, \ldots, p^n$, so by (b) $\prod_{k=0}^{n} \Phi_{p^k} = x^{p^n} - 1$, and also $\prod_{k=0}^{n-1} \Phi_{p^k} = x^{p^{n-1}} - 1$. Dividing,

$$\Phi_{p^n}(x) = \frac{x^{p^n}-1}{x^{p^{n-1}}-1} = \left.\frac{y^p-1}{y-1}\right|_{y=x^{p^{n-1}}} = \Phi_p\big(x^{p^{n-1}}\big).$$

(e) The primitive m-th roots of unity are $e^{2\pi i k/m}$ with $1 \leq k \leq m$ and $\gcd(k,m) = 1$, so $\deg \Phi_m = |(\mathbb{Z}/m\mathbb{Z})^*| =: \varphi(m)$. If $\gcd(m,n) = 1$, the Chinese Remainder Theorem gives a ring isomorphism $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$, which restricts to a bijection of unit groups $(\mathbb{Z}/mn\mathbb{Z})^* \cong (\mathbb{Z}/m\mathbb{Z})^* \times (\mathbb{Z}/n\mathbb{Z})^*$. Hence

$$\deg \Phi_{mn} = \deg \Phi_m \cdot \deg \Phi_n.$$

Problem 2

Galois groups and the Galois correspondence. Throughout, $\omega = e^{2\pi i/3}$, so $\omega^2 + \omega + 1 = 0$ and $\omega \notin \mathbb{R}$.

(a) $x^2 + 1$ **over** $\mathbb{R}$**.** Splitting field $\mathbb{C}$, $[\mathbb{C} : \mathbb{R}] = 2$, $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \{\mathrm{id}, \tau\} \cong \mathbb{Z}/2\mathbb{Z}$ with τ complex conjugation.

H	$\mathbb{C}^H$
$\{\mathrm{id}\}$	$\mathbb{C}$
$\mathrm{Gal}(\mathbb{C}/\mathbb{R})$	$\mathbb{R}$

(b) $x^3 - 1$ **over** $\mathbb{Q}$**.** $x^3 - 1 = (x-1)(x^2+x+1)$, splitting field $\mathbb{Q}(\omega)$ of degree 2 (as x^2+x+1 is irreducible over $\mathbb{Q}$). $\mathrm{Gal} = \{\mathrm{id}, \tau\} \cong \mathbb{Z}/2\mathbb{Z}$ with $\tau(\omega) = \omega^2$.

H	$\mathbb{Q}(\omega)^H$
$\{\mathrm{id}\}$	$\mathbb{Q}(\omega)$
$\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$	$\mathbb{Q}$

(c) $x^3 - 5$ **over** $\mathbb{Q}$**.** Let $\alpha = \sqrt[3]{5} \in \mathbb{R}$. The roots are $\alpha, \alpha\omega, \alpha\omega^2$, so the splitting field is $S = \mathbb{Q}(\alpha, \omega)$. Since $x^3 - 5$ is irreducible (Eisenstein at 5), $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, and $\omega \notin \mathbb{Q}(\alpha) \subseteq \mathbb{R}$ has degree 2 over $\mathbb{Q}(\alpha)$. So $[S : \mathbb{Q}] = 6 = |\mathrm{Gal}(S/\mathbb{Q})|$.

An automorphism is determined by $\alpha \mapsto$ one of 3 roots of $x^3 - 5$ and $\omega \mapsto \omega^{\pm 1}$; this gives 6 possibilities, so all occur. Define

$$\sigma : \ \alpha \mapsto \alpha\omega, \ \omega \mapsto \omega; \qquad \tau : \ \alpha \mapsto \alpha, \ \omega \mapsto \omega^2.$$

Then σ has order 3, τ has order 2, and $\tau\sigma\tau(\alpha) = \tau\sigma(\alpha) = \tau(\alpha\omega) = \alpha\omega^2 = \sigma^{-1}(\alpha)$, $\tau\sigma\tau(\omega) = \omega = \sigma^{-1}(\omega)$, so $\tau\sigma\tau = \sigma^{-1}$. Hence $\mathrm{Gal}(S/\mathbb{Q}) = \langle \sigma, \tau \rangle \cong S_3$.

Its six subgroups and fixed fields (composition $\sigma\tau = \sigma \circ \tau$):

- $\langle \tau \rangle$ fixes α: fixed field $\mathbb{Q}(\alpha)$.
- $\sigma\tau : \ \alpha \mapsto \alpha\omega, \ \omega \mapsto \omega^2$. Then $\sigma\tau(\alpha\omega^2) = \alpha\omega \cdot \omega^4 = \alpha\omega^2$: fixed field $\mathbb{Q}(\alpha\omega^2)$.
- $\sigma^2\tau : \ \alpha \mapsto \alpha\omega^2, \ \omega \mapsto \omega^2$. Then $\sigma^2\tau(\alpha\omega) = \alpha\omega^2 \cdot \omega^2 = \alpha\omega$: fixed field $\mathbb{Q}(\alpha\omega)$.
- $\langle \sigma \rangle$ fixes ω: fixed field $\mathbb{Q}(\omega)$.

(The three fields $\mathbb{Q}(\alpha), \mathbb{Q}(\alpha\omega), \mathbb{Q}(\alpha\omega^2)$ each have degree 3 over $\mathbb{Q}$, being generated by a root of $x^3 - 5$, matching the index 3 of an order-2 subgroup.)

H	S^H
$\{\mathrm{id}\}$	$\mathbb{Q}(\alpha, \omega)$
$\langle \tau \rangle$	$\mathbb{Q}(\alpha)$
$\langle \sigma\tau \rangle$	$\mathbb{Q}(\alpha\omega^2)$
$\langle \sigma^2\tau \rangle$	$\mathbb{Q}(\alpha\omega)$
$\langle \sigma \rangle$	$\mathbb{Q}(\omega)$
S_3	$\mathbb{Q}$

(d) $x^6 - 3x^3 + 2$ **over** $\mathbb{Q}$**.** With $y = x^3$ we get $y^2 - 3y + 2 = (y-1)(y-2)$, so $x^6 - 3x^3 + 2 = (x^3 - 1)(x^3 - 2)$. The roots are $1, \omega, \omega^2$ and $\beta, \beta\omega, \beta\omega^2$ with $\beta = \sqrt[3]{2}$. The splitting field is

$\mathbb{Q}(\beta,\omega)$, exactly the splitting field of x^3-2, of degree 6 (Problem 4(b) of HW2). Exactly as in (c) (replace 5 by 2): Gal $\cong S_3 = \langle\sigma,\tau\rangle$ with $\sigma(\beta)=\beta\omega,\ \sigma(\omega)=\omega,\ \tau(\beta)=\beta,\ \tau(\omega)=\omega^2$.

H	$\mathbb{Q}(\beta,\omega)^H$
$\{\mathrm{id}\}$	$\mathbb{Q}(\beta,\omega)$
$\langle\tau\rangle$	$\mathbb{Q}(\beta)$
$\langle\sigma\tau\rangle$	$\mathbb{Q}(\beta\omega^2)$
$\langle\sigma^2\tau\rangle$	$\mathbb{Q}(\beta\omega)$
$\langle\sigma\rangle$	$\mathbb{Q}(\omega)$
S_3	$\mathbb{Q}$

(e) x^5-1 **over** $\mathbb{Q}$**.** $x^5-1=(x-1)\Phi_5(x)$ with $\Phi_5=x^4+x^3+x^2+x+1$. Since $\Phi_5(x+1)=x^4+5x^3+10x^2+10x+5$ is Eisenstein at 5, Φ_5 is irreducible. With $\zeta=e^{2\pi i/5}$ the splitting field is $\mathbb{Q}(\zeta)$ of degree 4, and $\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})\cong(\mathbb{Z}/5\mathbb{Z})^*=\{\sigma_a:\zeta\mapsto\zeta^a\}$, cyclic of order 4 generated by σ_2 (the powers of 2 mod 5 are $2,4,3,1$).

Subgroups: $\{1\}$, $\{\sigma_1,\sigma_4\}=\langle\sigma_4\rangle$ (complex conjugation, $\zeta\mapsto\zeta^{-1}$), and the whole group. The fixed field of $\langle\sigma_4\rangle$ contains $\eta=\zeta+\zeta^{-1}=\zeta+\zeta^4$. Since $\eta+\eta'=-1$ and $\eta\eta'=-1$ for $\eta'=\zeta^2+\zeta^3$, η is a root of x^2+x-1, so $\eta=\frac{-1+\sqrt{5}}{2}$ (real, $=2\cos 72^\circ$) and $\mathbb{Q}(\eta)=\mathbb{Q}(\sqrt{5})$ has degree $2=[\mathbb{Q}(\zeta):\mathbb{Q}]/2$. Hence this is the fixed field.

H	$\mathbb{Q}(\zeta)^H$
$\{\mathrm{id}\}$	$\mathbb{Q}(\zeta)$
$\langle\sigma_4\rangle$	$\mathbb{Q}(\sqrt{5})=\mathbb{Q}(\zeta+\zeta^{-1})$
$\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})$	$\mathbb{Q}$

(f) x^6+x^3+1 **over** $\mathbb{Q}$**.** Since $x^9-1=(x^3-1)(x^6+x^3+1)$, we have $x^6+x^3+1=\Phi_9$, with roots the primitive 9-th roots of unity. Irreducibility: $\Phi_9(x+1)=x^6+6x^5+15x^4+21x^3+18x^2+9x+3$ is Eisenstein at 3. With $\zeta=e^{2\pi i/9}$ the splitting field is $\mathbb{Q}(\zeta)$ of degree 6, and Gal $\cong(\mathbb{Z}/9\mathbb{Z})^*=\{\sigma_a:\zeta\mapsto\zeta^a,\ a\in\{1,2,4,5,7,8\}\}$, cyclic of order 6 generated by σ_2 (powers of 2 mod 9: $2,4,8,7,5,1$).

Subgroups (orders $1,2,3,6$):

- $\langle\sigma_8\rangle=\{\sigma_1,\sigma_8\}$ (complex conjugation). Its fixed field contains $\zeta+\zeta^{-1}=2\cos 40^\circ$, and $[\mathbb{Q}(\zeta):\mathbb{Q}(\zeta+\zeta^{-1})]=2$ because ζ is a root of $X^2-(\zeta+\zeta^{-1})X+1$ and $\zeta\notin\mathbb{R}$. So the fixed field is $\mathbb{Q}(\zeta+\zeta^{-1})$, of degree 3.
- $\langle\sigma_4\rangle=\{\sigma_1,\sigma_4,\sigma_7\}$ (those $a\equiv 1 \bmod 3$). It fixes ζ^3, a primitive cube root of unity, so its fixed field contains $\mathbb{Q}(\zeta^3)=\mathbb{Q}(\omega)$, which has degree $2=6/3$. So the fixed field is $\mathbb{Q}(\omega)=\mathbb{Q}(\sqrt{-3})$.

H	$\mathbb{Q}(\zeta)^H$
$\{\mathrm{id}\}$	$\mathbb{Q}(\zeta)$
$\langle\sigma_8\rangle$ (order 2)	$\mathbb{Q}(\zeta+\zeta^{-1})$
$\langle\sigma_4\rangle$ (order 3)	$\mathbb{Q}(\zeta^3)=\mathbb{Q}(\sqrt{-3})$
$\mathrm{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q})\cong\mathbb{Z}/6\mathbb{Z}$	$\mathbb{Q}$

Problem 3

Prove that $\mathbb{Q}\big(\sqrt{2+\sqrt{2}}\big)$ is Galois over $\mathbb{Q}$ and find its Galois group.

Solution. Let $\alpha = \sqrt{2+\sqrt{2}} > 0$ and $\beta = \sqrt{2-\sqrt{2}} > 0$. From $\alpha^2 = 2+\sqrt{2}$ we get $(\alpha^2-2)^2 = 2$, i.e. $\alpha^4 - 4\alpha^2 + 2 = 0$. The polynomial $m(x) = x^4 - 4x^2 + 2$ is Eisenstein at 2, hence irreducible over $\mathbb{Q}$, so $[\mathbb{Q}(\alpha):\mathbb{Q}] = 4$. Its roots are $\pm\alpha, \pm\beta$ (indeed $\alpha^2, \beta^2 = 2 \pm \sqrt{2}$ are the roots of $y^2 - 4y + 2$).

Now $\sqrt{2} = \alpha^2 - 2 \in \mathbb{Q}(\alpha)$ and $\alpha\beta = \sqrt{(2+\sqrt{2})(2-\sqrt{2})} = \sqrt{2}$, so $\beta = \sqrt{2}/\alpha \in \mathbb{Q}(\alpha)$. Thus all roots $\pm\alpha, \pm\beta$ of m lie in $\mathbb{Q}(\alpha)$, so $\mathbb{Q}(\alpha)$ is the splitting field of m. Since $\operatorname{char}\mathbb{Q} = 0$, m is separable, and therefore $\mathbb{Q}(\alpha):\mathbb{Q}$ is Galois of degree 4, with $|\mathrm{Gal}| = 4$.

Since m is irreducible, there is $\sigma \in \mathrm{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q})$ with $\sigma(\alpha) = \beta$. Then

$$\sigma(\sqrt{2}) = \sigma(\alpha^2 - 2) = \beta^2 - 2 = -\sqrt{2}, \qquad \sigma(\beta) = \sigma\left(\frac{\sqrt{2}}{\alpha}\right) = \frac{-\sqrt{2}}{\beta} = -\alpha,$$

using $\sqrt{2}/\beta = \alpha$. Hence $\sigma:\ \alpha \mapsto \beta \mapsto -\alpha \mapsto -\beta \mapsto \alpha$ (because $\sigma(-\alpha) = -\beta$, $\sigma(-\beta) = \alpha$). So σ has order 4, and

$$\mathrm{Gal}\big(\mathbb{Q}(\sqrt{2+\sqrt{2}})/\mathbb{Q}\big) = \langle\sigma\rangle \cong \mathbb{Z}/4\mathbb{Z}.$$

(The unique intermediate field is the fixed field of $\sigma^2 : \alpha \mapsto -\alpha$, namely $\mathbb{Q}(\alpha^2) = \mathbb{Q}(\sqrt{2})$.) ■

Problem 4

Let $q = p^n$, $K = \mathbb{F}_q(t)$ and $f = x^q - x - t \in K[x]$, assumed irreducible over K. Find $\mathrm{Gal}(f/K)$.

Solution. Let E be a splitting field and $\alpha \in E$ a root of f, so $\alpha^q = \alpha + t$. For every $a \in \mathbb{F}_q$ we have $a^q = a$, hence

$$(\alpha + a)^q - (\alpha + a) - t = \alpha^q + a^q - \alpha - a - t = \alpha^q - \alpha - t = 0.$$

So $\alpha + a$ ($a \in \mathbb{F}_q$) are q distinct roots of f; as $\deg f = q$, these are all the roots. Therefore $E = K(\alpha)$, f is separable, and $E:K$ is Galois of degree q.

For each $a \in \mathbb{F}_q$, since f is irreducible there is a unique $\sigma_a \in \mathrm{Gal}(E/K)$ with $\sigma_a(\alpha) = \alpha + a$. Every automorphism is of this form (it sends α to a root of f), and $a \mapsto \sigma_a$ is a bijection $\mathbb{F}_q \to \mathrm{Gal}(E/K)$. It is a homomorphism: since σ_b fixes $a \in \mathbb{F}_q \subseteq K$,

$$\sigma_a\sigma_b(\alpha) = \sigma_a(\alpha + b) = \alpha + a + b = \sigma_{a+b}(\alpha).$$

Hence

$$\mathrm{Gal}(f/K) \cong (\mathbb{F}_{p^n}, +) \cong (\mathbb{Z}/p\mathbb{Z})^n,$$

an elementary abelian p-group of order p^n. ■

Problem 5

Let p be an odd prime, $K = \mathbb{F}_p(t)$, $f = x^4 - t \in K[x]$.

Preliminaries. f is Eisenstein at the prime t of the UFD $\mathbb{F}_p[t]$, so f is irreducible over K (Gauss). Since p is odd, $f' = 4x^3$ has only the root 0, which is not a root of f, so f is separable. Likewise $x^4 - 1$ is separable, so there are four fourth roots of unity $\{\pm 1, \pm i\}$ in $\overline{K}$, with $i^2 = -1$.

(a) Let α be a root of f (so $\alpha^4 = t$). If $c^4 = 1$ then $(c\alpha)^4 = t$, so $c\alpha$ is a root of f. Hence the roots of f are $\alpha, i\alpha, -\alpha, -i\alpha$, and the splitting field is $E = K(\alpha, i)$.

Case $p \equiv 1 \pmod 4$. Then $4 \mid p - 1 = |\mathbb{F}_p^*|$, and since $\mathbb{F}_p^*$ is cyclic it contains an element of order 4, i.e. $i \in \mathbb{F}_p \subseteq K$. Thus

$$E = K(\alpha), \qquad [E : K] = 4.$$

Case $p \equiv 3 \pmod 4$. Then $i \notin \mathbb{F}_p$ (as $4 \nmid p - 1$). Moreover $i \notin K$: if $i = u/v$ with $u, v \in \mathbb{F}_p[t]$ coprime, then $u^2 + v^2 = 0$ and so u, v have the same degree and comparing leading coefficients gives $-1 = (\mathrm{lc}(u)/\mathrm{lc}(v))^2$, a square in $\mathbb{F}_p$, contradicting $p \equiv 3 \pmod 4$. So $[K(i) : K] = 2$ and $K(i) = \mathbb{F}_{p^2}(t)$. Over $K(i) = \mathbb{F}_{p^2}(t)$, $f = x^4 - t$ is again Eisenstein at t in $\mathbb{F}_{p^2}[t]$, hence irreducible, so $[E : K(i)] = 4$. Therefore

$$E = K(\alpha, i), \qquad [E : K] = 8.$$

(b) *Case* $p \equiv 1 \pmod 4$. Here $i \in K$, and $\mathrm{Gal}(E/K)$ has order 4. Let $\sigma(\alpha) = i\alpha$ (this defines an automorphism since f is irreducible). Then $\sigma^k(\alpha) = i^k\alpha$, so σ has order 4 and

$$\mathrm{Gal}(E/K) = \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}.$$

Case $p \equiv 3 \pmod 4$. $|\mathrm{Gal}(E/K)| = 8$. An automorphism is determined by $\alpha \mapsto i^k\alpha$ ($k \in \mathbb{Z}/4$) and $i \mapsto \pm i$, giving 8 possibilities, all realized. Take

$$\sigma : \ \alpha \mapsto i\alpha, \ i \mapsto i; \qquad \tau : \ \alpha \mapsto \alpha, \ i \mapsto -i.$$

Then σ has order 4, τ has order 2, and $\tau\sigma\tau(\alpha) = \tau(i\alpha) = -i\alpha = \sigma^{-1}(\alpha)$, $\tau\sigma\tau(i) = \tau\sigma(-i) = \tau(-i) = i = \sigma^{-1}(i)$, so $\tau\sigma\tau = \sigma^{-1}$. Hence

$$\mathrm{Gal}(E/K) = \langle \sigma, \tau \mid \sigma^4 = \tau^2 = 1, \ \tau\sigma\tau = \sigma^{-1} \rangle \cong D_4 \text{ (dihedral of order 8)},$$

generated by σ and τ.

(c) Let $p \equiv 1 \pmod 4$, $E = K(\alpha)$, $G = \langle \sigma \rangle \cong \mathbb{Z}/4\mathbb{Z}$, $\sigma(\alpha) = i\alpha$. The subgroups of G are $\{1\}$, $\langle \sigma^2 \rangle$ and G. The element σ^2 sends $\alpha \mapsto -\alpha$ and thus fixes $\alpha^2 = \sqrt{t}$ (a square root of t), and $[K(\alpha^2) : K] = 2$. So:

H	E^H	$[E^H : K]$
$\{\mathrm{id}\}$	$E = K(\alpha) = K(\sqrt[4]{t})$	4
$\langle \sigma^2 \rangle$	$K(\alpha^2) = K(\sqrt{t})$	2
$\langle \sigma \rangle = G$	K	1

Problem 6

Let L be a finite field, $|L| = p^n$ with p prime (so $\mathrm{char}\, L = p$ and $\mathbb{F}_p \subseteq L$), and recall L^* is cyclic.

(a) Let θ generate L^*. Every nonzero element of L is a power of θ, so $L = \mathbb{F}_p(\theta) = \mathbb{F}_p[\theta]$. Let $g \in \mathbb{F}_p[x]$ be the minimal polynomial of θ over $\mathbb{F}_p$, an irreducible polynomial. The evaluation homomorphism $\mathbb{F}_p[x] \to L$, $x \mapsto \theta$, is surjective with kernel (g), so $L \cong \mathbb{F}_p[x]/(g(x))$.

(b) Put $q = p^n$. Since $|L^*| = q - 1$, every $y \in L^*$ satisfies $y^{q-1} = 1$, hence $y^q = y$ for all $y \in L$ (also for $y = 0$). So all q elements of L are roots of $x^q - x \in \mathbb{F}_p[x]$, which has degree q; hence

$$x^q - x = \prod_{y \in L} (x - y),$$

and L is the splitting field of $x^q - x$ over $\mathbb{F}_p$. The polynomial is separable ($(x^q - x)' = -1$). A splitting field of a separable polynomial is Galois, so $L : \mathbb{F}_p$ is Galois.

(c) *Uniqueness.* By (b), any field of cardinality p^n is a splitting field of $x^{p^n} - x$ over $\mathbb{F}_p$, and splitting fields are unique up to isomorphism.

Existence. Let S be a splitting field of $h = x^q - x$ over $\mathbb{F}_p$. h is separable, so it has q distinct roots in S. The set $R = \{y \in S : y^q = y\}$ of these roots is a subfield, because $y \mapsto y^q$ is a ring homomorphism (a power of the Frobenius $y \mapsto y^p$), so R is closed under addition and multiplication, contains $0, 1$, additive inverses and, for $y \neq 0$, $(y^{-1})^q = y^{-1}$. Since S is generated by the roots, $S = R$ and $|S| = q$. Hence there is exactly one field with p^n elements up to isomorphism, denoted $\mathbb{F}_{p^n}$.

(d) In characteristic p, $(a+b)^p = a^p + b^p$ (the binomial coefficients $\binom{p}{k}$, $0 < k < p$, are divisible by p), $(ab)^p = a^p b^p$ and $1^p = 1$, so $\varphi(y) = y^p$ is a ring homomorphism $\mathbb{F}_{p^n} \to \mathbb{F}_{p^n}$. It is injective (a ring homomorphism from a field is injective since $\ker \varphi$ is a proper ideal), hence bijective as the field is finite. Moreover φ fixes $\mathbb{F}_p$ (Fermat: $a^p = a$). So $\varphi \in \mathrm{Aut}(\mathbb{F}_{p^n}/\mathbb{F}_p)$.

(e) By (b), $\mathbb{F}_{p^n} : \mathbb{F}_p$ is Galois of degree $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$ (since $|\mathbb{F}_{p^n}| = p^n = p^{[\mathbb{F}_{p^n}:\mathbb{F}_p]}$), so $|\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)| = n$. We have $\varphi^k(y) = y^{p^k}$. For $1 \leq k < n$, $\varphi^k \neq \mathrm{id}$, since $x^{p^k} - x$ has at most $p^k < p^n$ roots and so cannot vanish on all of $\mathbb{F}_{p^n}$. On the other hand $\varphi^n(y) = y^{p^n} = y$. So φ has order exactly n, and therefore generates the whole Galois group:

$$\mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) = \langle \varphi \rangle \cong (\mathbb{Z}/n\mathbb{Z}, +).$$

Problem 7

Let $f_\ell = x^{2^\ell} + 1 \in \mathbb{F}_p[x]$.

(a) Let $g \in \mathbb{F}_p[x]$ have degree 2. If g is reducible it splits into linear factors over $\mathbb{F}_p$, hence over any extension. If g is irreducible, $F = \mathbb{F}_p[x]/(g)$ is a field with p^2 elements, hence $F \cong \mathbb{F}_{p^2}$ by Problem 6(c). In F, g has the root $\bar{x}$, so $g = (x - \bar{x})g_1$ with $\deg g_1 = 1$ and g splits in F. Thus g splits in $\mathbb{F}_{p^2}[x]$.

(b) For $p = 3$: $f_1 = x^2 + 1$ has no root in $\mathbb{F}_3$, since $0^2 + 1 = 1$, $1^2 + 1 = 2$, $2^2 + 1 = 5 = 2$ are all nonzero. A polynomial of degree 2 without a root is irreducible, so f_1 is irreducible over $\mathbb{F}_3$. Therefore

$$\mathbb{F}_9 = \mathbb{F}_3[x]/(x^2 + 1) = \{a + bi : \ a, b \in \mathbb{F}_3\}, \qquad i^2 = -1,$$

a field with 9 elements (unique up to isomorphism by 6(c)).

(c) $p = 2$. $f_\ell = (x + 1)^{2^\ell}$ and $x^{2^{\ell+1}} - 1 = (x + 1)^{2^{\ell+1}}$ both split over $\mathbb{F}_2$, so both splitting fields equal $\mathbb{F}_2$.

p odd. Work in an algebraic closure. Since $p \nmid 2^{\ell+1}$, $x^{2^{\ell+1}} - 1$ is separable, and its roots form a group μ of order $2^{\ell+1}$, which is cyclic (a finite subgroup of the multiplicative group of a

field). Let ζ generate μ. The splitting field of $x^{2^{\ell+1}}-1$ is $\mathbb{F}_p(\mu)=\mathbb{F}_p(\zeta)$. Since $f_\ell \mid x^{2^{\ell+1}}-1$, the roots of f_ℓ lie in μ, so the splitting field E_ℓ of f_ℓ satisfies $E_\ell \subseteq \mathbb{F}_p(\zeta)$. Conversely, ζ^{2^ℓ} has order 2 in μ, so $\zeta^{2^\ell}=-1$ (the only elements with $y^2=1$ are ± 1), i.e. $f_\ell(\zeta)=0$ and $\zeta \in E_\ell$. Thus $E_\ell=\mathbb{F}_p(\zeta)$, and the splitting fields of f_ℓ and $x^{2^{\ell+1}}-1$ coincide (in particular they are isomorphic).

(d) For $p=5$: $f_2=x^4+1$ and in $\mathbb{F}_5$ we have $1=-4$, so

$$x^4+1=x^4-4=(x^2-2)(x^2+2),$$

a product of two polynomials of degree 2. So f_2 is reducible over $\mathbb{F}_5$.

(e) **Claim:** for every $\ell \geq 2$ (e.g. $\ell=2$, $f_2=x^4+1$), f_ℓ is reducible in $\mathbb{F}_p[x]$ for every prime p.

Step 1: $(\mathbb{Z}/2^n\mathbb{Z})^*$ is not cyclic for $n \geq 3$. The elements -1, $2^{n-1}+1$, $2^{n-1}-1$ all square to 1 modulo 2^n: indeed $(2^{n-1}\pm 1)^2=2^{2n-2}\pm 2^n+1\equiv 1$ since $2n-2\geq n$. They are pairwise distinct and different from 1 modulo 2^n for $n\geq 3$ (e.g. $2^{n-1}+1\equiv -1$ would force $2^{n-1}=2$, i.e. $n=2$). So the group has at least three elements of order 2. A cyclic group has at most one element of order 2, hence $(\mathbb{Z}/2^n\mathbb{Z})^*$ is not cyclic.

Step 2. If $p=2$, $f_\ell=(x+1)^{2^\ell}$ is reducible ($\ell\geq 1$). Let p be odd and $\ell\geq 2$. Let ζ be a root of f_ℓ in an algebraic closure of $\mathbb{F}_p$. Then $\zeta^{2^\ell}=-1\neq 1$ and $\zeta^{2^{\ell+1}}=1$, so ζ has order exactly $2^{\ell+1}$. Let d be the order of p in the group $(\mathbb{Z}/2^{\ell+1}\mathbb{Z})^*$ of order 2^ℓ. By Step 1 (with $n=\ell+1\geq 3$) this group is not cyclic, so $d<2^\ell$. Now $2^{\ell+1}\mid p^d-1$, hence $\zeta^{p^d}=\zeta\cdot\zeta^{p^d-1}=\zeta$. So ζ is a root of $x^{p^d}-x$, i.e. $\zeta\in\mathbb{F}_{p^d}$ (the field of roots of $x^{p^d}-x$, by Problem 6), and consequently

$$\deg(\text{minimal polynomial of } \zeta \text{ over } \mathbb{F}_p)=[\mathbb{F}_p(\zeta):\mathbb{F}_p]\leq d<2^\ell=\deg f_\ell.$$

This minimal polynomial divides f_ℓ and is a proper factor of it, so f_ℓ is reducible over $\mathbb{F}_p$.

Hence $\ell=2$ (or any $\ell\geq 2$) works: x^4+1 is reducible over $\mathbb{F}_p$ for every prime p.